

Probability for Computer Science

① Counting

①. Counting Theory : Probability gets its genesis from counting theory.

How many unique outcomes are there? $\rightarrow$ Counting

Ex: Rolling a 6-sided die $\rightarrow$ 6 outcomes.

Get evens only $\rightarrow$ 3 outcomes.

Get odds only $\rightarrow$ 3 outcomes.

Roll two dice $\rightarrow$ 36 outcomes.

} These are all counting.

* Step Rule of Counting (Product Rule)

If an experiment has 2 steps $\rightarrow A \rightarrow B$ where $|A| = m$ & $|B| = n$, then $|A||B| = mn$.

Q: An individual pixel can have 256^3 colors (only 1 at a time).

Suppose you have three pixels, how many images can you generate using those three pixels?

$$\rightarrow 256^3 \times 256^3 \times 256^3 = 256^9$$

Steps: ① Choose color of pixel 1 $|P_1| = 256^3$

② " " " " 2 $|P_2| = 256^3$

③ " " " " 3 $|P_3| = 256^3$

$\therefore$ Total: 256^9

* The 'Sum' rule of counting (OR)

→ Only if $|A \text{ and } B| = 0$ ($A \cap B = \emptyset$) (Mutual Exclusivity)

Q: A 6 bit string is sent over a network. The valid set of strings recognised by the receiver must either start with "01" or end with "10". How many possible strings are there?

$$\rightarrow \frac{1}{2} \frac{1}{2} \frac{1}{2} \frac{1}{2} \frac{1}{2} \frac{1}{2} = 2^4 \quad \text{or} \quad 01 \text{ --- } = 2^4$$

But, $01 \frac{1}{2} \frac{1}{2} 10 = 2^2 \rightarrow$ these are counted twice

So, total possible strings $2^4 + 2^4 - 2^2$

$$N = |A| + |B| - |A \cap B|$$